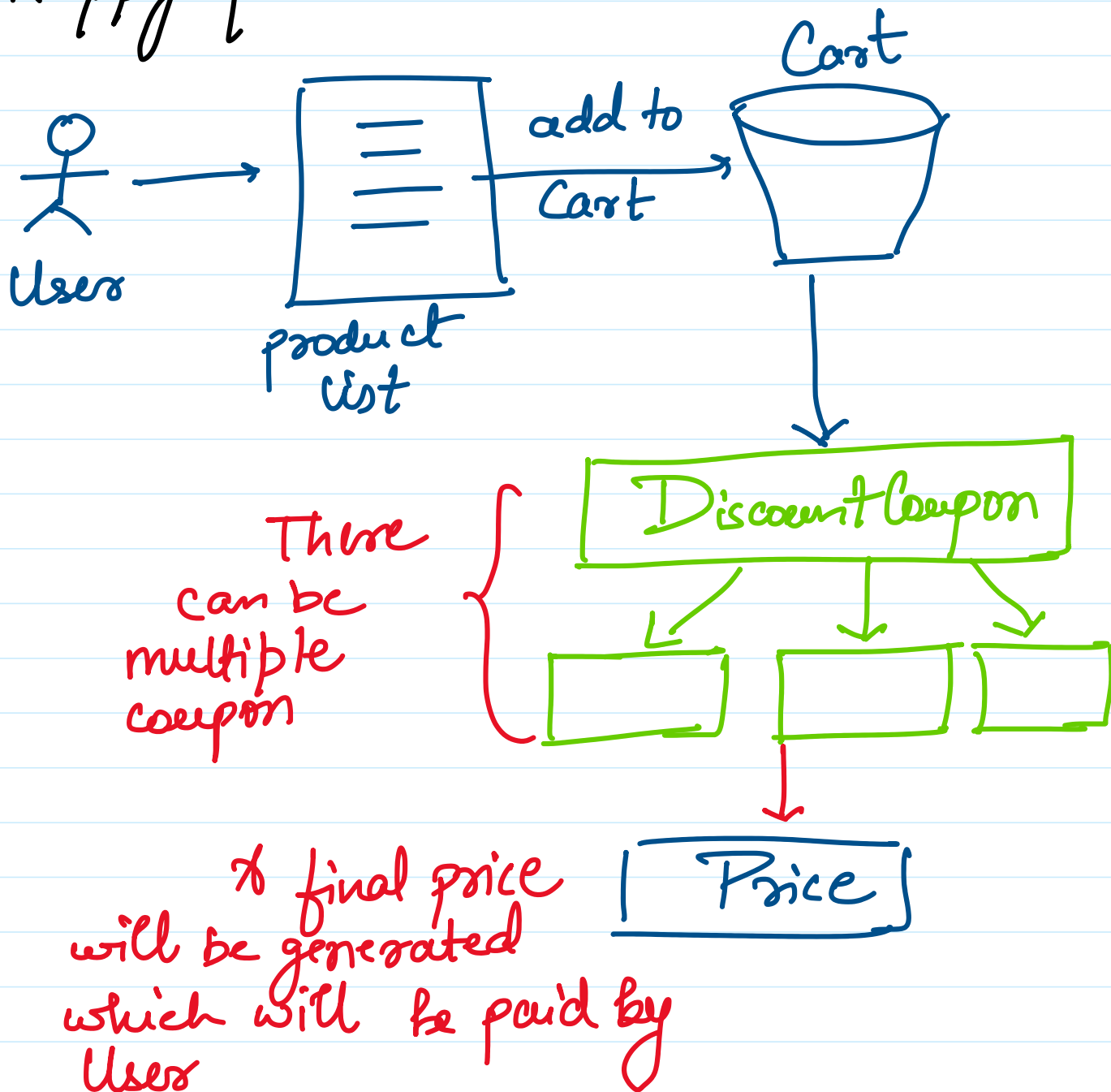


functional Requirements

1. We can add new Coupons at Runtime.
2. Both Cart level & product level discount
3. Support Multiple Coupon types.
(Seasonal offer, loyalty Discount, Banking Discount.)
4. Support both flat & %age discount
— Scalable like 20% of 120₹
5. One Coupon can/cannot be applied on top of other coupon
6. Thread safe.
↳ for example we have only 4 coupon & only first 4 customers can take benefit of it.
↓
Multithreading
↓
locking Mechanism.

Happy flow

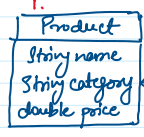


Plug - an - play - Our design

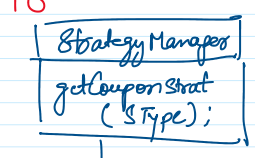
1. Start with creating a 'product' class

Here we are taking a generic item of 'Inventory Management System' or you can take any other type of item. For simplicity we are taking 'Product' class which can have any item.

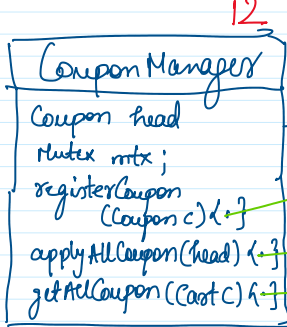
Here we will add 'CartItem' in cart not 'Product' b/c same product can be added multiple times



boolean value to give loyalty discount to loyalty members
logically it should be fetched from 'User' class but we are not making User class
it will internally create 'CartItem' using p & q and then add it in 'items'
This subtract 'amt' from 'originalTotal' & save it in 'finalTotal'



Response core for



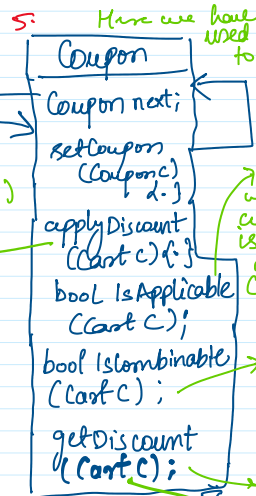
12

(1...*)

add coupon at last of head

calls head.applyDiscount()

Get all Coupon that can be applied on Cart C

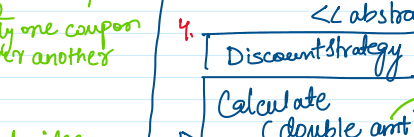


5. Here we have used COR to apply one coupon over another

This decides whether the current coupon is applicable on this 'Cart C' or not

This decides whether other coupons are applicable above it or not

This will get the discount amount on this Cart 'C'



4. it take the 'amt' and calculate the discounted amount. How it calculates? Depends upon the class

return (discount Amt)

all this will override the abstract method.

6.

7.

8.

9.

Banking Coupon

Loyalty Coupon

Bulk Purchase Coupon

Seasonal Coupon

getDiscount (Cart C)

How getDiscount (Cart C) will work?

Suppose 'Banking Coupon' will have some 'Strategy' 'ds'

ds.calculate (amt)

This 'amt' will come from Cart 'C' 'originalTotal'

This is like a 'Template method Pattern'

```

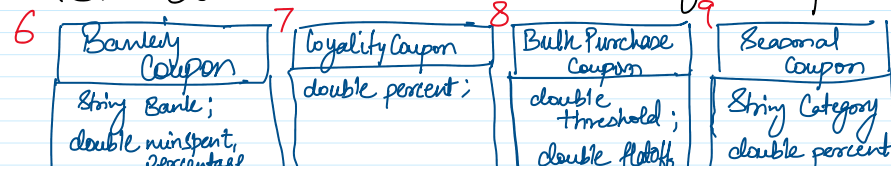
if (curr != null) {
    if (curr.isApplicable()) {
        x = curr.getDiscount();
        c.applyDiscount(x);
    }
}

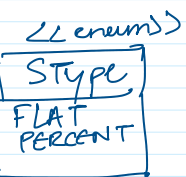
```

apply Dis

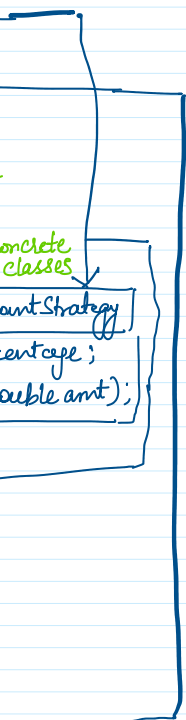
This pro repeat for pointer if

let's create the detailed class of all coupons





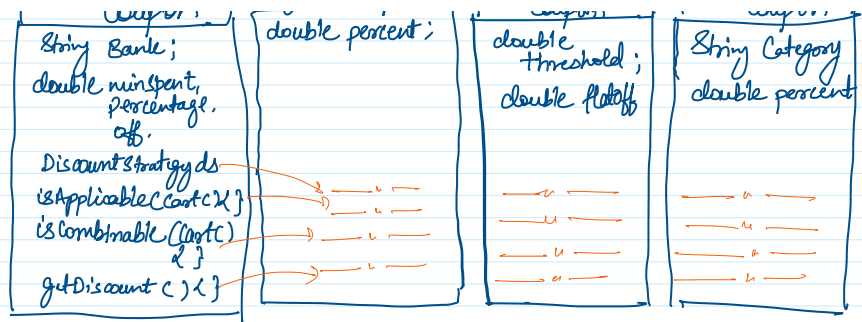
possible to
 create DiscountStrategy
 Coupons



}

count()

process will
 'next'
 'it is not
 NULL.



Clean UML Diagram
 → See Notes

Code → See Notes.

